TOPU KUMAR MONDOL

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EDUCATION

Rajshahi University of Engineering & Technology (RUET) — **Rajshahi, Bangladesh**
B.Sc. in Electrical & Computer Engineering (ECE) — *Expected Graduation: Oct 2028*

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Systems

PROJECTS

OpsIQ — Multi-Mode AI Assistant for Chat, Document Q&A and Incident Analysis — **GitHub**
Python · FastAPI · LangGraph · LangChain · FAISS · PostgreSQL · SQLAlchemy — *2026*

- Built a session-based AI assistant supporting general chat, FAISS-based document Q&A and log-triggered incident analysis, using 3 LLM instances configured at temperatures of 0.7, 0.3 and 0.1 respectively, with responses streamed through Server-Sent Events.
- Generated postmortems through a 5-node LangGraph pipeline with parallel log analysis and timeline extraction, followed by report-grounded chat; added 19 tests checking expected concepts in synthetic scenarios based on published GitLab, Cloudflare and AWS postmortems.
- Implemented session recovery using PostgreSQL for conversation summaries, recent messages and reports, plus disk-backed FAISS indexes; automated testing and deployment through a Docker-based GitHub Actions CI/CD pipeline with a 123-test suite.

Private Search — Privacy-Focused Metasearch Engine (In Progress) — **GitHub**
Python · FastAPI · asyncio · httpx · Pydantic · Docker · GitHub Actions · Locust — *2026*

- Handled ~1,250 concurrent users at p95 under 2 s in Locust load tests (simulated providers) by querying multiple search providers in parallel with async FastAPI, with per-source timeouts so one slow provider never blocks the results.
- Kept users anonymous to search providers, with zero search queries or IP addresses logged, by routing every search through the server and forwarding only the query, backed by 100+ Pytest tests and a Docker + GitHub Actions CI/CD pipeline.

ContentPlatform — Personalized Recommendation Engine — **GitHub**
FastAPI · PostgreSQL · SQLAlchemy 2.0 · JWT + OTP Auth · APScheduler · Docker · Pytest · Locust — *2025*

- Built a personalized recommendation engine combining weighted interactions, exponential time decay, category limits and softmax exploration; refreshed preferences every 50 minutes using batched SQL aggregation and PostgreSQL bulk upserts.
- Load-tested with Locust at up to 500 concurrent users, recording 100% request success at 100 users and a 6.39% failure rate at 500 users; diagnosed SQLAlchemy connection pool exhaustion from server logs.
- Enforced account blocks on the next authenticated request rather than waiting for the default 7-day JWT expiry; automated deployment to Hugging Face Spaces through a GitHub Actions CI/CD pipeline gated by 58 tests.

TECHNICAL SKILLS

Languages: Python, C++, C, SQL

AI / ML: LangChain, LangGraph, RAG, Retrieval Systems, Vector Databases (FAISS), Embeddings, LLM Integration

Backend & Databases: FastAPI, asyncio, httpx, SQLAlchemy 2.0, PostgreSQL, REST APIs, JWT Auth, APScheduler

Cloud & Infrastructure: Docker, CI/CD (GitHub Actions), Render, Hugging Face Spaces, Neon (managed PostgreSQL)

Tools: Git, GitHub, Pytest, Locust, Pydantic

CERTIFICATIONS

Machine Learning Specialization — **Dec 2025**
DeepLearning.AI / Andrew Ng (Coursera) · Credential ID: PTP0CHLR5O6S

CS50x: Introduction to Computer Science — **Oct 2024**
HarvardX (edX)

CS50P: Introduction to Programming with Python — **2024**
HarvardX (edX)

ADDITIONAL

- Solved 400+ data structures and algorithms problems across LeetCode, GeeksforGeeks and Codeforces — spanning arrays, trees, graphs, binary search and dynamic programming